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United States Patent	12403532
Kind Code	B2
Date of Patent	September 02, 2025
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Cutting member, holding device and cutting tool with arrangement for indicating rotational alignment between cutting member and holding device

Abstract

A cutting tool includes a cutting member having a shank portion located in a clamping recess of a clamping sleeve adaptor of a holding device. The shank portion has a peripheral shank surface including a switching arrangement including two circumferentially spaced apart circumferential outer portions and a circumferential inner portion extending therebetween. The circumferential inner portion has opposite electrical conductivity to the two circumferential outer portions. The holding device has an alignment indicating unit including an electric switch including two spaced apart conductive switch members configured to be in electrically conductive communication or non-communication with each other thereby closing or opening the electric switch, so that an electric circuit established or broken, forming first and second states of the cutting tool, respectively. Switching between the first and second states is realized by rotation of the cutting member with respect to the clamping sleeve adaptor.

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Appl. No.:	17/220099
Filed:	April 01, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220314334 A1	Oct. 06, 2022

Publication Classification

Int. Cl.: B23B25/06 (20060101); B23Q17/22 (20060101)

U.S. Cl.:

CPC B23B25/06 (20130101); B23Q17/2225 (20130101);

Field of Classification Search

CPC: B23B (25/06); B23Q (17/2225); B23Q (17/2216); B23Q (2716/008); B23Q (2717/003); B23Q (17/006); B23Q (17/0909); B23Q (17/0919); B23Q (17/0957); Y10T (82/2572); F21V (33/0084)

USPC: 33/639; 408/16

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Background/Summary

FIELD OF THE INVENTION

(1) The subject matter of the present application relates to cutting tools having a cutting member releasably clamped in a recess formed in a clamping sleeve of a holding device, and further in particular to such cutting tools having an electronic means for indicating rotational alignment of the cutting member with respect to the holding device.

BACKGROUND OF THE INVENTION

(2) Cutting tools can include a holding device having a clamping sleeve and a cutting member having a cylindrical shank portion releasably clamped in a clamping recess located on the clamping sleeve. Prior to clamping, the cutting tool can have a means for indicating rotational alignment of the cutting member with respect to the holding device (specifically, the clamping sleeve). Rotational alignment of the cutting member with respect to the holding device is realized when the height of cutting edge is positioned so that it engages with the center of the work-piece.

(3) An example of such a cutting tool is disclosed in, for example, US2002/0035902A1, having a cutting tool system which includes a clamping device and a cutting tool mounted therein. The cutting tool carries a cutting edge and includes a shaft clamped within an aperture of the clamping device. Prior to being clamped, the shaft is movable relative to the clamping device in either a longitudinal or rotational direction, in order to locate the cutting edge in a desired position. Either the clamping device or the cutting tool carries a spring-loaded device which engages a recess

formed in the other of the cutting tool and clamping device when the cutting tool and its cutting edge are in a desired position, to provide an indication that the cutting tool is in such desired position by providing a sudden increase in a force necessary to displace the cutting tool out of such desired position.

(4) Another example is disclosed in U.S. Pat. No. 10,254,104 B2, having an apparatus and method for center height alignment of a boring bar. The apparatus includes a seat member with a bottom and a back, and a digital angle gauge mounted to the seat member. The method includes the steps of: assembling an apparatus comprising a seat member and a digital angle gauge; calibrating the apparatus such that a Display of the digital angle gauge reads zero degrees; and aligning the center height of the boring bar by rotating the boring bar such that the Display of the digital angle gauge reads zero degrees when the apparatus is mounted onto the boring bar.

(5) Yet another example is disclosed in U.S. Pat. No. 10,717,133 B2, having a cutting tool, and a turning machine including the cutting tool. The cutting tool includes a tool bar extending along an axis, a cutting head located at the tool bar, and at least one sensor integrated with the tool bar or the cutting head. A rotational orientation of the cutting tool with respect to the axis is estimated based on output provided by the at least one sensor. In at least some embodiments, the at least one sensor includes accelerometers configured to measure acceleration in at least two directions.

SUMMARY OF THE INVENTION

(6) In accordance with a first aspect of the subject matter of the present application there is provided a holding device comprising: a clamping sleeve adaptor, extending along a sleeve central axis, and an alignment indicating unit located at the clamping sleeve adaptor; wherein: the clamping sleeve adaptor comprises: a radially outward facing sleeve peripheral surface extending circumferentially about the sleeve central axis, the sleeve peripheral surface intersecting, and forming a boundary of, a forward sleeve end surface at a forward sleeve end of the clamping sleeve adaptor; and a clamping recess, extending along a recess central axis, the clamping recess comprising a recess peripheral surface which extends circumferentially about the recess central axis and which opens out to the forward sleeve end surface; and the alignment indicating unit comprises an alignment indicating unit housing and an electric circuit located therewithin, the electric circuit comprising: a power supply unit; an indicator; and an electric switch, for turning the indicator on and off, the electric switch comprising two spaced apart conductive switch members protruding from the alignment indicating unit housing towards the recess central axis, the switch members configured to be in electrically conductive communication or non-communication with each other, thereby closing or opening the electric switch so that the electric circuit is established or broken, respectively.

(7) In accordance with a second aspect of the subject matter of the present application there is provided A cutting tool comprising: a holding device of the type describe above; and a cutting member, having a cutting member longitudinal axis defining opposite forward to rearward directions and opposite rotational preceding and succeeding directions, the cutting member comprising: a cutting portion comprising a cutting edge; a shank portion formed axially opposite, and rearward of, the cutting portion; and two axially opposite cutting member end surfaces and a cutting member peripheral surface extending therebetween and about the cutting member longitudinal axis, the cutting member peripheral surface at the shank portion being cylindrical and forming a peripheral shank surface; and a switching arrangement located at the peripheral shank surface and comprising two circumferentially spaced apart circumferential outer portions and a circumferential inner portion extending therebetween, the circumferential inner portion having opposite electrical conductivity to the two circumferential outer portions; wherein: the shank portion is located in the clamping recess of the holding device.

(8) In accordance with a third aspect of the subject matter of the present application there is provided A cutting member, having a cutting member longitudinal axis defining opposite forward to rearward directions and opposite rotational preceding and succeeding directions, the cutting

member comprising: a cutting portion comprising a cutting edge; a shank portion formed axially opposite, and rearward of, the cutting portion; and two axially opposite cutting member end surfaces and a cutting member peripheral surface extending therebetween and about the cutting member longitudinal axis, the cutting member peripheral surface at the shank portion being cylindrical and forming a peripheral shank surface; and a switching arrangement located at the peripheral shank surface and comprising two circumferentially spaced apart circumferential outer portions and a circumferential inner portion extending therebetween, the circumferential inner portion having opposite electrical conductivity to the two circumferential outer portions; and a conductive substrate and a non-conductive external element formed on part of the conductive substrate; wherein: the conductive substrate forms the circumferential inner portion and the non-conductive external element forms the two circumferential outer portions, or the conductive substrate forms the two circumferential outer portions and the non-conductive external element forms the circumferential inner portion.

(9) It is understood that the above-said is a summary, and that features described hereinafter may be applicable in any combination to the subject matter of the present application, for example, any of the following features may be applicable to the holding device, cutting tool or cutting member:

(10) The recess peripheral surface can have a cylindrical shape.

(11) The two conductive switch members can be located outside the clamping recess.

(12) The recess peripheral surface can open out to the forward sleeve end surface at a clamping recess opening. In a front view of the holding device, the two conductive switch members can extend, in a direction towards the recess central axis, at least to the clamping recess opening.

(13) The two conductive switch members can be spring loaded.

(14) The alignment indicating unit can be releasably attached to the clamping sleeve adaptor.

(15) The clamping sleeve adaptor can comprise a sleeve flange adjacent the forward sleeve end surface. The alignment indicating unit can be releasably attached to the clamping sleeve adaptor at the sleeve flange.

(16) The indicator can comprise an indicating light.

(17) Each of the two conductive switch members can have a switch member contact surface subtending a switch member contact angle at the recess central axis.

(18) The two conductive switch members can be axially spaced apart along the recess central axis by a switch member axial distance. The two conductive switch members can be circumferentially aligned about the recess central axis.

(19) The two conductive switch members can be angularly spaced apart about the recess central axis by a switch member circumferential angle.

(20) The cutting tool can be adjustable between a first state and a second state.

(21) In the first state of the cutting tool: the cutting member can be oriented in a first rotated position about the cutting member longitudinal axis; the two conductive switch members can be electrically connected to one another by whichever one of the inner and two circumferential outer portions is conductive so that the two conductive switch members are in electrically conductive communication with each other; and the electric switch can be closed, the electric circuit can be established and the indicator can be on.

(22) In the second state of the cutting tool: the cutting member can be oriented in a second rotated position about the cutting member longitudinal axis, the first and second rotated positions being different from each other; at least one of the two conductive switch members can be spaced apart from whichever one of the inner and two circumferential outer portions is conductive so that the two conductive switch members are not in electrically conductive communication with each other; and the electric switch can be open, the electric circuit can be broken and the indicator (84) can be off.

(23) The circumferential inner portion can be conductive and the two circumferential outer portions can be non-conductive. In the first state of the cutting tool, the two switch members can both

contact the circumferential inner portion. The first state of the cutting tool can indicate rotational alignment between the cutting member and the holding device.

(24) The circumferential inner portion can be non-conductive and the two circumferential outer portions can be conductive. In the first state of the cutting tool, the two switch members can both contact one of the two circumferential outer portions. The first state of the cutting tool (20) can indicate rotational non-alignment between the cutting member and the holding device.

(25) The first state of the cutting tool can be realized when the cutting member is rotationally oriented within a first state tolerance angle with respect to the clamping sleeve adaptor about the cutting member longitudinal axis.

(26) The two conductive switch members can be circumferentially aligned about the recess central axis. Each of the two conductive switch members can have a switch member contact surface subtending a switch member contact angle at the recess central axis. The circumferential inner portion can subtend an inner portion angle at the cutting member longitudinal axis. The first state tolerance angle can have a value equal to the switch member contact angle plus twice the inner portion angle.

(27) The two conductive switch members can be angularly spaced apart about the recess central axis by a switch member circumferential angle. The circumferential inner portion can subtend an inner portion angle at the cutting member longitudinal axis. The switch member circumferential angle can be less than the inner portion angle. The first state tolerance angle can have a value equal to the inner portion angle minus the switch member circumferential angle.

(28) In the second state of the cutting tool, at least one of the two conductive switch members can be in entire contact with whichever one of the inner and two circumferential outer portions is non-conductive.

(29) The circumferential inner portion can be conductive and the two circumferential outer portions can be non-conductive.

(30) The cutting tool can further comprise a conductive substrate and a non-conductive external element formed on part of the conductive substrate. The conductive substrate can form the circumferential inner portion and the non-conductive external element can form the two circumferential outer portions or the conductive substrate can form the two circumferential outer portions and the non-conductive external element can form the circumferential inner portion.

(31) The non-conductive external element can be a coated layer formed on part of the conductive substrate.

(32) The conductive substrate can be formed from steel.

(33) The peripheral shank surface can comprise a peripheral groove. The non-conductive external element can be located in the peripheral groove.

(34) The circumferential inner portion can be elongated and extend in a direction from one of cutting member end surfaces towards the other.

(35) The circumferential inner portion can extend parallel to the cutting member longitudinal axis.

(36) The circumferential inner portion may not extend to the cutting member end surface located at the shank portion.

(37) The circumferential inner portion can subtend an inner portion angle at the cutting member longitudinal axis. The inner portion angle can be greater than or equal to 5° and less than or equal to 10° .

(38) The cutting member can comprise an additional switching arrangement located at the peripheral shank surface, such that the cutting member has two switching arrangements which are rotationally offset from each other about the cutting member longitudinal axis.

(39) The two switching arrangements can be rotationally offset from each other about the cutting member longitudinal axis by 180° .

(40) The two circumferential outer portions can comprise a rotationally preceding circumferential outer portion extending from the circumferential inner portion in the rotationally preceding

direction and a rotationally succeeding circumferential outer portion extending from the circumferential inner portion in the rotationally succeeding direction. The rotationally preceding circumferential outer portion and the rotationally succeeding circumferential outer portion can intersect each other distal the circumferential inner portion.

(41) The cutting edge can be formed at the intersection of a rake surface and a relief surface. The rotationally preceding direction can have a tangential component direction at the cutting edge, the rake surface facing generally in the tangential component direction. The cutting member can have a longitudinal plane containing the cutting member longitudinal axis and intersecting the cutting edge. The switching arrangement can be located on the side of the longitudinal plane that extends therefrom in the tangential component direction.

(42) The cutting member can be releasably clamped in the clamping recess of the clamping sleeve adaptor of the holding device. The alignment indicating unit can be detached from the clamping sleeve adaptor.

(43) the conductive substrate can form the circumferential inner portion and the non-conductive external element can form the two circumferential outer portions.

(44) The non-conductive external element can be a coated layer formed on part of the conductive substrate.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) For a better understanding of the present application and to show how the same may be carried out in practice, reference will now be made to the accompanying drawings, in which:

(2) FIG. 1 is a perspective view of a cutting tool in accordance with a first embodiment of the present application;

(3) FIG. 2 is an exploded perspective view of the cutting tool shown in FIG. 1;

(4) FIG. 3 is a perspective view of a cutting member shown in FIGS. 1 and 2;

(5) FIG. 4 is a top view of the cutting member shown in FIG. 3;

(6) FIG. 5 is a cross-sectional view of the cutting member taken along the line V-V in FIG. 4, with a cutting insert superimposed;

(7) FIG. 6 is a front view of a holding device shown in FIGS. 1 and 2;

(8) FIG. 7 is a perspective view of a clamping sleeve adaptor shown in FIG. 6;

(9) FIG. 8 is a top view of the clamping sleeve adaptor shown in FIG. 7;

(10) FIG. 9 is a side view of the clamping sleeve adaptor shown in FIG. 7;

(11) FIG. 10 is a front view of the clamping sleeve adaptor shown in FIG. 7;

(12) FIG. 11 is a front view of an alignment indicating unit shown in FIGS. 1 and 2;

(13) FIG. 12 is a cross-sectional view of the alignment indicating unit taken along the line XII-XII in FIG. 11;

(14) FIG. 13 is a schematic cross-sectional view of two conductive switch members and a switching arrangement, in a radial plane therethrough, showing two opposing threshold positions, in accordance with a first embodiment of the present application; and

(15) FIG. 14 is a schematic cross-sectional view of two conductive switch members and a switching arrangement, in a radial plane therethrough, showing two opposing threshold positions, in accordance with a second embodiment of the present application.

(16) It will be appreciated that for simplicity and clarity of illustration, elements shown in the figures have not necessarily been drawn to scale. For example, the dimensions of some of the elements may be exaggerated relative to other elements for clarity, or several physical components may be included in one functional block or element. Where considered appropriate, reference numerals may be repeated among the figures to indicate corresponding or analogous elements.

DETAILED DESCRIPTION OF THE INVENTION

(17) In the following description, various aspects of the subject matter of the present application will be described. For purposes of explanation, specific configurations and details are set forth in sufficient detail to provide a thorough understanding of the subject matter of the present application. However, it will also be apparent to one skilled in the art that the subject matter of the present application can be practiced without the specific configurations and details presented herein.

(18) Attention is first drawn to FIGS. **1** and **2** showing a cutting tool **20**, for chip removal, in accordance with embodiments of the subject matter of the present application. The cutting tool **20** can be a turning tool, being fixed while a work-piece rotates during cutting operations.

(19) Referring in particular to FIG. **3**, the cutting tool **20** includes a cutting member **22**. The cutting member **22** has a cutting member longitudinal axis A which defines opposite forward and rearward directions D.sub.F, D.sub.R. It should be noted that use of the terms “forward” and “rearward” throughout the description and claims refer to a relative position in a direction of the cutting member longitudinal axis A towards the left and right, respectively, in FIG. **4**. In the present disclosure, “forward” is associated with the cutting end of the rotary cutting tool **20**. It should also be noted that use of the terms “axial” and “radial” throughout the description and claims are with respect to the cutting member longitudinal axis A, unless otherwise stated. The cutting member longitudinal axis A also defines opposite rotational preceding and succeeding directions D.sub.P, D.sub.S. It should further be noted that use of the terms “rotational preceding” and “rotational succeeding” throughout the description and claims refer to a relative position in a direction of the cutting member longitudinal axis A clockwise and counter-clockwise, respectively, in FIG. **5**.

(20) As seen in FIGS. **3** and **4**, the cutting member **22** includes cutting portion **24**. The cutting portion **24** includes a cutting edge **26**, designed for metal cutting. The cutting edge **26** is formed at the intersection of a rake surface **28** and a relief surface **30**. The rotationally preceding direction D.sub.P has a tangential component direction T at the cutting edge **26**. Referring to FIG. **5**, in accordance with some embodiments of the subject matter of the present application, the rake surface **28** can face generally in the tangential component direction T. The relief surface **30** can face generally in the forward direction D.sub.F. The cutting member **22** has a longitudinal plane P containing the cutting member longitudinal axis A and intersecting the cutting edge **26**.

(21) The cutting member **22** includes a shank portion **32** formed axially opposite, and rearward of, the cutting portion **24**. That is to say, the cutting member **22** extends from the shank portion **32** to the cutting portion **24** along the cutting member longitudinal axis A. Part of the shank portion **32** is designed to be clamped by a clamping sleeve adaptor, as described later in the description.

(22) It is noted that, in this non-limiting example shown in the drawings, the cutting member **22** can include an insert holder **34** having a holder pocket **36** at the cutting portion **24**. A cutting insert **38** can be releasably retained in the holder pocket **36**.

(23) The cutting member **22** includes two axially opposite cutting member end surfaces **40a**, **40b** and a cutting member peripheral surface **42** extending therebetween and about the cutting member longitudinal axis A. The cutting member longitudinal axis A intersects the two cutting member end surfaces **40a**, **40b**. One of the two cutting member end surfaces **40a** (a forward cutting member end surface) is located at the cutting portion **24** and the other one of the two cutting member end surfaces **40b** (a rear cutting member end surface) is located at the shank portion **32**. The cutting member peripheral surface **42** at the shank portion **32** is cylindrical and forms a peripheral shank surface **44**. In accordance with some embodiments of the subject matter of the present application, the peripheral shank surface **44** can be devoid of flat surfaces which are designed to be clampingly engaged by clamping screws.

(24) The cutting member **22** includes an electrical switching arrangement **46** located at the peripheral shank surface **44**. The switching arrangement **46** is designed to provide a means for closing and opening an electric switch located on a separate alignment indicating unit, as described

later in the description. As shown in FIG. 5, in accordance with some embodiments of the subject matter of the present application, the switching arrangement **46** can be located on the side of the longitudinal plane P that extends therefrom in the tangential component direction T.

(25) The switching arrangement **46** includes two circumferentially spaced apart, circumferential outer portions **48a**, **48b** and a circumferential inner portion **50** which extends therebetween. The two circumferential outer portions **48a**, **48b** are spaced apart from each other in the opposite rotational preceding and succeeding directions D.sub.P, D.sub.S. At the surface of the cutting member, the circumferential inner portion **50** has an inner portion axial length d1 along the cutting member longitudinal axis A and an inner portion circumferential width w1 in a direction around the cutting member peripheral surface **42**. The circumferential outer portions **48a**, **48b** and the circumferential inner portion **50** are located at the peripheral shank surface **44**. In accordance with some embodiments of the subject matter of the present application, the circumferential inner portion **50** can be generally radially level with the two circumferential outer portions **48a**, **48b**. That is to say, the circumferential inner portion **50** can transition smoothly and continuously with the two circumferential outer portions **48a**, **48b**. Thus, the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** can lay on the same cylindrical surface as the peripheral shank surface **44**. In particular, there may be no stepped arrangement at their intersection forming a recessed region for engaging with, for example, a sphere or cylinder, as disclosed in US2002/0035902A1.

(26) In a switching arrangement **46**, two circumferential outer portions **48a**, **48b** include a rotationally preceding circumferential outer portion **48a** which extends from the circumferential inner portion **50** in the rotationally preceding direction D.sub.P and a rotationally succeeding circumferential outer portion **48b** which extends from the circumferential inner portion **50** in the rotationally succeeding direction D.sub.S.

(27) In accordance with some embodiments of the subject matter of the present application, the cutting member **22** can include a single switching arrangement **46**. In such a configuration, the rotationally preceding circumferential outer portion **48a** and the rotationally preceding circumferential outer portion **48b** can intersect each other distal (e.g. opposite) the circumferential inner portion **50**. The rotationally preceding circumferential outer portion **48a** and the rotationally succeeding circumferential outer portion **48b** can merge smoothly and continuously at their intersection.

(28) In accordance with some other embodiments of the subject matter of the present application, the cutting member **22** can include an additional switching arrangement **46** located at the **46** located at the peripheral shank surface **44**, such that the cutting member **22** includes two switching arrangements **46** which can be rotationally offset from each other about the cutting member longitudinal axis A. In such a configuration, the rotationally preceding circumferential outer portion **48a** of a first switching arrangement **46** and the rotationally succeeding circumferential outer portion **48b** of the additional, second switching arrangement **46** can intersect each other (optionally smoothly and continuously) distal both circumferential inner portions **50**. The two switching arrangements **46** can be rotationally offset from each other about the cutting member longitudinal axis A by 180°, as see in FIG. 5.

(29) As seen best in FIG. 4, in accordance with some embodiments of the subject matter of the present application, the circumferential inner portion **50** can be elongated along the cutting member longitudinal axis A and thus extend in a direction from one of cutting member end surfaces **40a**, **40b** towards the other. The circumferential inner portion **50** can extend parallel to the cutting member longitudinal axis A. The circumferential inner portion **50** may not extend to the cutting member end surface **40a** located at the cutting portion **24**. The circumferential inner portion **50** may not extend to the cutting member end surface **40b** located at the shank portion **32**. In other words, the circumferential inner portion **50** is axially spaced apart from both cutting member end surfaces **40a**, **40b**. Referring to FIG. 5, the circumferential inner portion **50** can subtend an inner portion

angle α at the cutting member longitudinal axis A. The inner portion angle α can be greater than or equal to 5° and less than or equal to 10° . Preferably, the inner portion angle α can be greater than or equal to 6.5° and less than or equal to 8.5° .

(30) The circumferential inner portion **50** has opposite electrical conductivity to the two circumferential outer portions **48a**, **48b**. That is to say, either the circumferential inner portion **50** is electrically conductive and the two circumferential outer portions **48a**, **48b** are both non-conductive (i.e. electrical insulators), or the circumferential inner portion **50** is non-conductive and the two circumferential outer portions **48a**, **48b** are both conductive. Stated differently, when the circumferential inner portion **50** is conductive, the two circumferential outer portions **48a**, **48b** are non-conductive and vice versa. Thus, the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** are neither all conductive nor all non-conductive in a single configuration. In accordance with some embodiments of the subject matter of the present application the circumferential inner portion **50** can be conductive and the two circumferential outer portions **48a**, **48b** can be non-conductive. In accordance with some other embodiments of the subject matter of the present application the circumferential inner portion **50** can be non-conductive and the two circumferential outer portions **48a**, **48b** can be conductive. It is understood that the terms “conductivity”, “non-conductivity” etc., used in the description, refers to the electrical type of conductivity.

(31) Referring again to FIG. 5, in accordance with some embodiments of the subject matter of the present application, the cutting member **22** can include a conductive substrate **52**. The conductive substrate **52** can be formed from steel. The cutting member **22** can include a non-conductive external element **54** formed on part of the conductive substrate **52**. The non-conductive external element **54** can transition smoothly with the conductive substrate **52**. The non-conductive external element **54** can be non-fluid. The non-conductive external element **54** can be solid. The non-conductive external element **54** can be deformable. For example, the non-conductive external element **54** can be formed from rubber or plastic. In accordance with some other embodiments of the subject matter of the present application, the non-conductive external element **54** can be a coated layer formed on part of the conductive substrate **52**.

(32) In accordance with some embodiments of the subject matter of the present application, the conductive substrate **52** can form the circumferential inner portion **50** and the non-conductive external element **54** can form the two circumferential outer portions **48a**, **48b**. Alternatively, in accordance with some other embodiments of the subject matter of the present application, the conductive substrate **52** can form the two circumferential outer portions **48a**, **48b** and the non-conductive external element **54** can form the circumferential inner portion **50**.

(33) In accordance with some embodiments of the subject matter of the present application, the peripheral shank surface **44** can include a peripheral groove **56**. The non-conductive external element **54** can be located in the peripheral groove **56**. The non-conductive external element **54** can fill the peripheral groove **56** so as to be flush with the peripheral shank surface **44**. In certain other configurations, the non-conductive external element **54** can extend out of the peripheral groove **56**. As seen in FIG. 5, in a cross-sectional view of the cutting member taken in a plane perpendicular to the cutting member longitudinal axis A, the peripheral groove **56** has a substantially semicircular shape, which intersects the cutting member peripheral surface **42** at a generally 90° angle.

(34) Referring to FIG. 6, the cutting tool **20** also has a holding device **58**. As shown in FIGS. 7-10, the holding device **58** includes a clamping sleeve adaptor **60**. The clamping sleeve adaptor **60** has a sleeve central axis B along which it extends. The clamping sleeve adaptor **60** is typically made from steel. The clamping sleeve adaptor **60** includes a radially outward facing sleeve peripheral surface **62** extending circumferentially about the sleeve central axis B. The sleeve peripheral surface **62** intersects, and forms a boundary of, a forward sleeve end surface **64a** at a forward sleeve end **61a** of the clamping sleeve adaptor **60**. The clamping sleeve adaptor **60** includes a rear sleeve end surface **64b** at a rear sleeve end **61b** of the clamping sleeve adaptor **60**, the rear sleeve

end **61b** being axially opposite the forward sleeve end **61a** with respect to the sleeve central axis B, and rearward of the forward sleeve end **61a**. As seen in FIG. 8, in accordance with some embodiments of the subject matter of the present application, clamping sleeve adaptor **60** can include a sleeve flange **66** adjacent the forward sleeve end surface **64a**.

(35) The clamping sleeve adaptor **60** includes a clamping recess **68**. The clamping recess **68** serves to receive the cutting member **22**. The clamping recess **68** has a recess central axis C, along which it extends. The recess central axis C can be parallel to, and optionally co-incident with, the sleeve central axis B. The clamping recess **68** is recessed in the forward sleeve end surface **64a**. The clamping recess **68** includes a recess peripheral surface **70** which extends circumferentially about the recess central axis C and which opens out to the forward sleeve end surface **64a** at a clamping recess opening **72**. The recess peripheral surface **70** is designed to clampingly engage the cutting member **22**. In accordance with some embodiments of the subject matter of the present application, the recess peripheral surface **70** can have a shape that corresponds to the cutting member **22** at the shank portion **32**. Thus, the recess peripheral surface **70** can have a cylindrical shape. The recess peripheral surface **70** can open out to the rear sleeve end surface **64b**. Thus, the clamping recess **68** can be a through recess. The clamping sleeve adaptor **60** can include a sleeve slit **74** which opens out to the recess peripheral surface **70** and the sleeve peripheral surface **62** from the forward sleeve end surface **64a** to the rear sleeve end surface **64b**. The sleeve slit **74** allows the radial dimension of the clamping recess **68** to be reduced in order to clamp the cutting member **22**. It is noted that there may be no through hole which opens out to the recess peripheral surface **70** and the sleeve peripheral surface **62**, for receiving a spring-loaded device, as disclosed in US2002/0035902A1.

(36) Referring to FIGS. 6 and 11-12, the holding device **58** includes an electronic module in the form of an alignment indicating unit **76** located at the clamping sleeve adaptor **60**. In some embodiments (not shown), the alignment indicating unit **76** is non-releasably attached to the clamping sleeve adaptor **60**. In the embodiment seen in e.g., FIGS. 1 and 2, the alignment indicating unit **76** is releasably attached to clamping sleeve adaptor **60** and therefore may be detached therefrom. In accordance with some embodiments of the subject matter of the present application, the alignment indicating unit **76** can be releasably attached to the clamping sleeve adaptor **60**, preferably at a forward end thereof. The alignment indicating unit **76** can be attached to the clamping sleeve adaptor **60** after the shank portion **32** has been inserted in the clamping recess **68**. The alignment indicating unit **76** can be detached from the clamping sleeve adaptor **60** prior to metal-cutting. Advantageously, such removal can prevent damage to the alignment indicating unit **76** during cutting operations. Moreover, chip flow is not obstructed by the alignment indicating unit **76**. Further moreover, coolant channel openings (not shown) at the forward sleeve end surface **64a** would not be obstructed by the alignment indicating unit **76**. In certain configurations, the alignment indicating unit **76** can be releasably attached to the clamping sleeve adaptor **60** at the sleeve flange **66**.

(37) The alignment indicating unit **76** includes an alignment indicating unit housing **78**. The alignment indicating unit **76** also includes an electric circuit **80** located within the alignment indicating unit housing **78**. The electric circuit **80** includes a power supply unit **82**. The power supply unit **82** provides power to the electric circuit **80**. The power supply unit **82** is preferably a battery though other power sources may also be used.

(38) The electric circuit **80** also includes an indicator **84**. The indicator **84** acts to provide an indication to a user that the cutting member **22** is rotationally aligned with respect to clamping sleeve adaptor **60**. The indicator **84** can provide a visual indication. For example, the indicator **84** can include an indicating light **86**. However, in other embodiments, an audible indicator may be used instead of, or in addition to, a visual indicator.

(39) The electric circuit **80** also includes an electric switch **88**. The electric circuit **80** can be established (i.e. the electric circuit **80** can be completed) or broken when the electric switch **88** is closed or opened, respectively, in order to operate the indicator **84**. The electric switch **88** includes

two spaced apart conductive switch members **90a**, **90b**. The two conductive switch members **90a**, **90b** protrude from the alignment indicating unit housing **78** towards the recess central axis C. The two conductive switch members **90a**, **90b** are configured to be in electrically conductive communication or non-communication with each other thereby closing or opening the electric switch **88**, respectively.

(40) In accordance with some embodiments of the subject matter of the present application, the two conductive switch members **90a**, **90b** are located outside the clamping recess **68**. In a front view of the holding device **58**, the two conductive switch members **90a**, **90b** can extend, in a direction towards the recess central axis C, at least to the clamping recess opening **72**. In the same view, the two conductive switch members **90a**, **90b** can overlap the clamping recess opening **72**. The two conductive switch members **90a**, **90b** can be pins, elongated in a direction towards the recess central axis C. Referring to FIG. **12**, in accordance with some embodiments of the subject matter of the present application, the two conductive switch members **90a**, **90b** can be axially spaced apart along the recess central axis C by a switch member axial distance D.

(41) In accordance with some embodiments of the subject matter of the present application, the two conductive switch members **90a**, **90b** can be circumferentially aligned about recess central axis C. In accordance with some other embodiments of the subject matter of the present application, the two conductive switch members **90a**, **90b** can be angularly spaced apart about the recess central axis C by a switch member circumferential angle β .

(42) In accordance with some embodiments of the subject matter of the present application, the two conductive switch members **90a**, **90b** can be spring loaded. The two conductive switch members **90a**, **90b** can be identical. Each of the two conductive switch members **90a**, **90b** can have a switch member contact surface **92**, for contact with the two circumferential outer portions **48a**, **48b** and the circumferential inner portion **50**. The switch member contact surface **92** subtends a switch member contact angle γ at the recess central axis C. Specifically, the switch member contact angle γ is subtended by circumferential extremities of the switch member contact surface **92**. The switch member contact surface **92** can have a shape complementary to the two circumferential outer portions **48a**, **48b** and the circumferential inner portion **50**. Alternatively, the switch member contact surface **92** can be convexly curved. For example, the switch member contact surface **92** can have a frusto-spherical shape. Contact between the switch member contact surface **92** and the two circumferential outer portions **48a**, **48b** and the circumferential inner portion **50** can be point-contact.

(43) Reference is now reverted to FIGS. **1-2**, showing the cutting tool **20**, in accordance with the subject matter of a third aspect of the present application. The cutting tool **20** includes the cutting member **22** releasably clamped in the clamping recess **68** of the clamping sleeve adaptor **60** of the holding device **58** forming an assembled position of the cutting tool **20**. In accordance with some embodiments of the subject matter of the present application, the sleeve central axis B can be co-incident with the cutting member longitudinal axis A.

(44) To realize the assembled position of the cutting tool **20**, the shank portion **32** is inserted in the clamping recess **68**. After insertion, the shank portion **32** is located in the clamping recess **68** of the holding device **58**. From such a configuration, and prior to realizing the assembled position of the cutting tool **20**, the cutting tool **20** can be adjustable between a first state in which the indicator light is on, and a second state in which the indicator light is off.

(45) In the first state of the cutting tool **20**, the cutting member **22** is oriented in a first rotated position about the cutting member longitudinal axis A. The two conductive switch members **90a**, **90b** are electrically connected to one another (i.e. are in electrically conductive communication with each other) by whichever one of the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** is conductive, thereby closing the electric switch **88** so that the electric circuit **80** is established. Upon the electric circuit **80** being established, the indicator **84** is turned on.

(46) In the second state of the cutting tool **20**, the cutting member **22** is oriented in a second rotated position about the cutting member longitudinal axis A. The first and second rotated to positions being different from each other. The two conductive switch members **90a**, **90b** are not electrically connected to one another, the electric switch **88** is open, and the indicator light is off.

(47) In embodiments of a first cutting member type in which the circumferential inner portion **50** is conductive and the two circumferential outer portions **48a**, **48b** are non-conductive, in the first state of the cutting tool **20**, the two switch members **90a**, **90b** both contact the circumferential inner portion **50**. A closed electric circuit **88** (i.e., the first state) indicates that the cutting member **22** is rotationally aligned with respect to the clamping sleeve adaptor **60**. At such a stage, the cutting member **22** can be tightly clamped into the clamping recess **68**, thereby being ready for metal cutting operations.

(48) Furthermore, in embodiments of the first cutting member type, the first state of the cutting tool **20** can be realized when the cutting member **22** is rotationally oriented within a first state tolerance angle θ with respect to the clamping sleeve adaptor **60** about the cutting member longitudinal axis A. The first state tolerance angle θ can be subtended by first and second arc end points E1, E2 of a first state arc F at the cutting member longitudinal axis A. The value of the first state tolerance angle θ determines the angular range over which the indicator **84** indicates either rotational alignment or non-alignment between the cutting member **22** and the holding device **58** (i.e. specifically, the clamping sleeve adaptor **60**).

(49) In contrast, in embodiments of a second cutting member type in which the circumferential inner portion **50** is non-conductive and the two circumferential outer portions **48a**, **48b** are conductive, in the first state of the cutting tool **20**, the two switch members **90a**, **90b** both contact one of the two circumferential outer portions **48a**, **48b**. A closed electric circuit **88** (i.e., the first state) indicates that the cutting member **22** is rotationally non-aligned with respect to the clamping sleeve adaptor **60**.

(50) Reference is now made to FIGS. **13** and **14** representing schematic cross-sectional views of two conductive switch members and a switching arrangement, in a radial plane through two conductive switch members and the switching arrangement, showing two opposing threshold positions. The two opposing threshold positions represent position where the cutting tool **20** transforms from the first state to the second state when the cutting member **22** is rotated in the rotational preceding and succeeding directions D.sub.P, D.sub.S, respectively. In particular, it is noted that the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** are depicted as straight lines in these figures, when in reality they are defined by a radius.

Furthermore, in FIGS. **13** and **14**, the conductive switch members **90a**, **90b** are shown as having cross-sections of constant switch member circumferential width w_2 and constant axial spacing D, along their heights, rather than having tapering proximate their contact surface **92**, as seen in FIG. **12**.

(51) FIG. **13** shows the configuration where the two conductive switch members **90a**, **90b** are circumferentially aligned about the recess central axis C (so only the forwardmost one is visible). The first arc end point E1 (i.e. the leftmost arc end point) is defined by a rotationally preceding-most point (i.e. leading point in the rotational preceding direction D.sub.P) on the circumferential inner portion **50'** in a first position at the moment the circumferential inner portion **50'** loses contact with the two conductive switch members **90a**, **90b** as the cutting member **22** is rotated in the rotational preceding direction D.sub.P. The second arc end point E2 (i.e. the rightmost arc end point) is defined by a rotationally succeeding-most point (i.e. a leading point in the rotational succeeding direction D.sub.S) on the circumferential inner portion **50''** in a second position at the moment the circumferential inner portion **50''** loses contact with the two conductive switch members **90a**, **90b** as the cutting member **22** is rotated in the rotational succeeding direction D.sub.S. The first state tolerance angle θ can have a value equal to the switch member contact angle γ plus twice the inner portion angle α . In FIG. **13**, at the surface of the cutting member **22**, the

switch member circumferential width w_2 of the two conducting switch members **90a**, **90b** is more important in contacting the circumferential inner portion **50** to close the switch, than the switch member axial distance D (as shown in FIG. **12**) between the two conducting switch members **90a**, **90b**. This is because the axial lengths of the circumferential inner and outer portions is several times longer than the switch member axial distance D .

(52) Referring to FIG. **14**, in order to make the alignment indication more precise the first state tolerance angle θ can be reduced. An example of such a configuration is one having the two conductive switch members **90a**, **90b** angularly spaced apart (i.e., circumferentially spaced apart) about the recess central axis C by a switch member circumferential angle β . The switch member circumferential angle β can be less than the inner portion angle α . In FIG. **14**, at the surface of the cutting member **22**, the circumferentially spaced apart switch members **90a**, **90b** are spaced apart from one another by a switch member circumferential distance D_1 while the circumferential inner portion **50** again has the inner portion circumferential width w_1 . It can be seen from FIG. **14** that, at the surface of the cutting member **22**, the inner portion circumferential width w_1 of the to circumferential inner portion **50** must be greater than the switch member circumferential distance D_1 between the two conductive switch members **90a**, **90b**, i.e., $w_1 > D_1$.

(53) Still referring to FIG. **14**, the first arc end point E_1 (i.e. the leftmost arc end point) is defined by a rotationally preceding-most point (i.e. a leading point in the rotational preceding direction $D_{sub.P}$) on the circumferential inner portion **50'** in the first position at the moment the circumferential inner portion **50'** loses contact with the conductive switch member **90b** that leads in the rotational succeeding direction $D_{sub.S}$, as the cutting member **22** is rotated in the rotational preceding direction $D_{sub.P}$. The second arc end point E_2 (i.e. the rightmost arc end point) is defined by a rotationally preceding-most point (i.e. a leading point in the rotational preceding direction $D_{sub.P}$ —that is, the same point as used for the first arc end point E_1) on the circumferential inner portion **50''** in the second position at the moment the circumferential inner portion **50''** loses contact with the conductive switch member **90a** that leads in the rotational preceding direction $D_{sub.P}$, as the cutting member **22** is rotated in the rotational succeeding direction $D_{sub.S}$. It is noted that the circumferential inner portion **50'**, **50''** shown in the first and second positions overlap each other (unlike in FIG. **13**). The first state tolerance angle θ can have a value equal to the inner portion angle α minus the switch member circumferential angle β . It is noted that in this configuration, the first state tolerance angle θ is not dependent on the switch member contact angle γ . In FIG. **14**, in a first cutting member type, the switch member circumferential width w_2 is not as important in completing the circuit by contacting the circumferential inner portion **50**, as compared to the axial spacing D between the two conducting switch members **90a**, **90b**.

(54) In accordance with some embodiments of the subject matter of the present application, the second state of the cutting tool **20** can be realized by rotating the cutting member **22** about the cutting member longitudinal axis A with respect to the clamping sleeve adaptor **60** (i.e. in the rotational preceding or succeeding directions $D_{sub.P}$, $D_{sub.S}$. In the second state of the cutting tool **20**, at least one of the two conductive switch members **90a**, **90b** can be spaced apart from whichever one of the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** is conductive. i.e. the two conductive switch members **90a**, **90b** are not electrically connected to one another by whichever one of the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** is conductive so that the two conductive switch members **90a**, **90b** are not in electrically conductive communication with each other, thereby opening the electric switch **88** so that the electric circuit **80** is broken. Upon the electric circuit **80** being broken, the indicator **84** can be turned off, indicating that the cutting member **22** is rotationally misaligned with respect to clamping sleeve adaptor **60**. At least one of the two conductive switch members **90a**, **90b** can be in entire contact with whichever one of the circumferential inner portion **50** and the two circumferential outer portions **48a**, **48b** is non-conductive.

(55) Although the subject matter of the present application has been described to a certain degree of particularity, it should be understood that various alterations and modifications could be made without departing from the spirit or scope of the invention as hereinafter claimed. For example, the cutting member 22 can be integrally formed with the shank portion 32 having a one-piece construction with cutting portion 24. The cutting member 22 can be made from cemented carbide. Moreover, in the configuration where the circumferential inner portion 50 is non-conductive, the non-conductive external element 54 can be formed by an air-gap.

Claims

1. A holding device (58) comprising: a clamping sleeve adaptor (60), extending along a sleeve central axis (B), and an alignment indicator (76) attached to the clamping sleeve adaptor (60); wherein: the clamping sleeve adaptor (60) comprises: a radially outward facing sleeve peripheral surface (62) extending circumferentially about the sleeve central axis (B), the sleeve peripheral surface (62) intersecting, and forming a boundary of, a forward sleeve end surface (64a) at a forward sleeve end (61a) of the clamping sleeve adaptor (60); and a clamping recess (68), extending along a recess central axis (C), the clamping recess (68) comprising a recess peripheral surface (70) which extends circumferentially about the recess central axis (C) and which opens out to the forward sleeve end surface (64a); and the alignment indicator (76) comprises an alignment indicator housing (78) and an electric circuit (80) located therewithin, the electric circuit (80) comprising: a power supply (82); an indicator (84) having an on state configured to indicate a shank portion of a cutting member being received and rotationally aligned in the clamping recess and an off state; and an electric switch (88) having a closed configuration corresponding to the on state of the indicator (84) and an open configuration corresponding to the off state of the indicator (84), the electric switch (88) comprising two spaced apart conductive switch members (90a, 90b) protruding from and through the alignment indicator housing (78) towards the recess central axis (C), each of the two switch members (90a, 90b) having a switch member contact surface (92) which is formed on a part of the switch member (90a, 90b) that protrudes from the alignment indicator housing (78), in the closed configuration of the electric switch (88) a first one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a first position and a second one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a second position such that the switch member contact surfaces are in electrically conductive communication with one another through the shank portion, in the open configuration of the electric switch (88) the first one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a third position and the second one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a fourth position such that the switch member contact surfaces are not in electrical communication with one another through the shank portion.

2. The holding device (58), according to claim 1, wherein the recess peripheral surface (70) has a cylindrical shape.

3. The holding device (58), according to claim 1, wherein the two conductive switch members (90a, 90b) are located outside the clamping recess (68).

4. The holding device (58), according to claim 1, wherein: the recess peripheral surface (70) opens out to the forward sleeve end surface (64a) at a clamping recess opening (72); and in a front view of the holding device (58), the two conductive switch members (90a, 90b) extend, in a direction towards the recess central axis (C), at least to the clamping recess opening (72).

5. The holding device (58), according to claim 1, wherein the two conductive switch members (90a, 90b) are urged away from the alignment indicator (76).

6. The holding device (58), according to claim 1, wherein the alignment indicator (76) is releasably

attached to the clamping sleeve adaptor (60).

7. The holding device (58), according to claim 1, wherein: the clamping sleeve adaptor (60) comprises a sleeve flange (66) adjacent the forward sleeve end surface (64a); and the alignment indicator (76) is releasably attached to the clamping sleeve adaptor (60) at the sleeve flange (66).

8. The holding device (58), according to claim 1, wherein the indicator (84) comprises an indicating light (86).

9. The holding device (58), according to claim 1, wherein each switch member contact surface (92) subtends at a switch member contact angle (γ) at the recess central axis (C).

10. The holding device (58), according to claim 1, wherein: the two conductive switch members (90a, 90b) are axially spaced apart along the recess central axis (C) by a switch member axial distance (D); and the two conductive switch members (90a, 90b) are circumferentially aligned about the recess central axis (C).

11. The holding device (58), according to claim 1, wherein the two conductive switch members (90a, 90b) are angularly spaced apart about the recess central axis (C) by a switch member circumferential angle (β).

12. The holding device (58), according to claim 1, wherein the recess peripheral surface (70) has a shape configured to correspond to the shank portion of the cutting member.

13. A cutting tool (20) comprising: a holding device (58) comprising: a clamping sleeve adaptor (60), extending along a sleeve central axis (B), and an alignment indicator (76) attached to the clamping sleeve adaptor (60); the clamping sleeve adaptor (60) comprising: a radially outward facing sleeve peripheral surface (62) extending circumferentially about the sleeve central axis (B), the sleeve peripheral surface (62) intersecting, and forming a boundary of, a forward sleeve end surface (64a) at a forward sleeve end (61a) of the clamping sleeve adaptor (60); and a clamping recess (68), extending along a recess central axis (C), the clamping recess (68) comprising a recess peripheral surface (70) which extends circumferentially about the recess central axis (C) and which opens out to the forward sleeve end surface (64a); and the alignment indicator (76) comprises an alignment indicator housing (78) and an electric circuit (80) located therewithin, the electric circuit (80) comprising: a power supply (82); an indicator (84) having an on state configured to indicate a shank portion of a cutting member being received and rotationally aligned in the clamping recess and an off state; and an electric switch (88) having a closed configuration corresponding to the on state of the indicator (84) and an open configuration corresponding to the off state of the indicator (84), the electric switch (88) comprising two spaced apart conductive switch members (90a, 90b) protruding from the alignment indicator housing (78) towards the recess central axis (C), each of the two switch members (90a, 90b) having a switch member contact surface (92) which is formed on a part of the switch member (90a, 90b) that protrudes from the alignment indicator housing (78), in the closed configuration of the electric switch (88) a first one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a first position and a second one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a second position such that the switch member contact surfaces are in electrically conductive communication with one another through the shank portion, in the open configuration of the electric switch (88) the first one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a third position and the second one of the switch member contact surfaces is configured to contact the shank portion received in the clamping recess at a fourth position such that the switch member contact surfaces are not in electrical communication with one another through the shank portion; the cutting member (22), having a cutting member longitudinal axis (A) defining opposite forward to rearward directions (D.sub.F, D.sub.R) and opposite rotational preceding and succeeding directions (D.sub.P, D.sub.S), the cutting member (22) comprising: a cutting portion (24) comprising a cutting edge (26); the shank portion (32) formed axially opposite, and rearward of, the cutting portion (24); and two axially opposite cutting member end surfaces (40a, 40b) and a cutting member peripheral

surface (42) extending therebetween and about the cutting member longitudinal axis (A), the cutting member peripheral surface (42) at the shank portion (32) being cylindrical and forming a peripheral shank surface (44); and a switching arrangement (46) located at the peripheral shank surface (44) and comprising two circumferentially spaced apart circumferential outer portions (48a, 48b) and a circumferential inner portion (50) extending therebetween, one of either the circumferential inner portion (50) or the two circumferential outer portions (48a, 48b) being conductive and the other one of either the circumferential inner portion (50) or the two circumferential outer portions (48a, 48b) being non-conductive; wherein: the shank portion (32) is located in the clamping recess (68) of the holding device (58).

14. The cutting tool (20), according to claim 13, wherein: the cutting tool (20) is adjustable between a first state and a second state; in the first state of the cutting tool (20): the cutting member (22) is oriented in a first rotated position about the cutting member longitudinal axis (A); the switch member contact surfaces (92) are electrically connected to one another by whichever one of the inner and two circumferential outer portions (48a, 48b) is conductive so that the two conductive switch members (90a, 90b) are in electrically conductive communication with each other; and the electric switch (88) is closed, the electric circuit (80) is established and the indicator (84) is on; and in the second state of the cutting tool (20): the cutting member (22) is oriented in a second rotated position about the cutting member longitudinal axis (A), the first and second rotated positions being different from each other; at least one of the two conductive switch members (90a, 90b) is spaced apart from whichever one of the inner and two circumferential outer portions (48a, 48b) is conductive so that the switch member contact surfaces (92) are not electrically connected to one another so that the two conductive switch members (90a, 90b) are not in electrically conductive communication with each other; and the electric switch (88) is open, the electric circuit (80) is broken and the indicator (84) is off.

15. The cutting tool (20), according to claim 14, wherein: the circumferential inner portion (50) is conductive and the two circumferential outer portions (48a, 48b) are non-conductive; in the first state of the cutting tool (20), the two switch members (90a, 90b) both contact the circumferential inner portion (50); and the first state of the cutting tool (20) indicates rotational alignment between the cutting member (22) and the holding device (58).

16. The cutting tool (20), according to claim 14, wherein: the circumferential inner portion (50) is non-conductive and the two circumferential outer portions (48a, 48b) are conductive; in the first state of the cutting tool (20), the two switch members (90a, 90b) both contact one of the two circumferential outer portions (48a, 48b); and the first state of the cutting tool (20) indicates rotational non-alignment between the cutting member (22) and the holding device (58).

17. The cutting tool (20), according to claim 14, wherein: the two conductive switch members (90a, 90b) are angularly spaced apart about the recess central axis (C) by a switch member circumferential angle (β); the circumferential inner portion (50) subtends an inner portion angle (α) at the cutting member longitudinal axis (A); the switch member circumferential angle (β) is less than the inner portion angle (α); and the first state tolerance angle (θ) has a value equal to the inner portion angle (α) minus the switch member circumferential angle (β).

18. The cutting tool (20), according to claim 14, wherein: in the second state of the cutting tool (20): at least one of the two conductive switch members (90a, 90b) is in entire contact with whichever one of the inner and two circumferential outer portions (48a, 48b) is non-conductive.

19. The cutting tool (20), according to claim 13, wherein: the circumferential inner portion (50) is conductive and the two circumferential outer portions (48a, 48b) are non-conductive.

20. The cutting tool (20), according to claim 13, further comprising a conductive substrate (52) and a non-conductive external element (54) formed on part of the conductive substrate (52) wherein: the conductive substrate (52) forms the circumferential inner portion (50) and the non-conductive external element (54) forms the two circumferential outer portions (48a, 48b); or the conductive substrate (52) forms the two circumferential outer portions (48a, 48b) and the non-conductive

external element (54) forms the circumferential inner portion (50).

21. The cutting tool (20), according to claim 20, wherein the non-conductive external element (54) is a coated layer formed on part of the conductive substrate (52).

22. The cutting tool (20), according to claim 20, wherein the conductive substrate (52) is formed from steel.

23. The cutting tool (20), according to claim 20, wherein: the peripheral shank surface (44) comprises a peripheral groove (56); and the non-conductive external element (54) located in the peripheral groove (56).

24. The cutting tool (20), according to claim 13, wherein the circumferential inner portion (50) is elongated and extends in a direction from one of cutting member end surfaces (40a, 40b) towards the other.

25. The cutting tool (20), according to claim 24, wherein the circumferential inner portion (50) extends parallel to the cutting member longitudinal axis (A).

26. The cutting tool (20), according to claim 24, wherein the circumferential inner portion (50) does not extend to the cutting member end surface (40b) located at the shank portion (32).

27. The cutting tool (20), according to claim 13, wherein: the circumferential inner portion (50) subtends an inner portion angle (α) at the cutting member longitudinal axis (A); and the inner portion angle (α) is greater than or equal to 5° and less than or equal to 10° .

28. The cutting tool (20), according to claim 13, wherein the cutting member (22) comprises an additional switching arrangement (46) located at the peripheral shank surface (44), such that the cutting member (22) has two switching arrangements (46) which are rotationally offset from each other about the cutting member longitudinal axis (A).

29. The cutting tool (20), according to claim 28, wherein the two switching arrangements (46) are rotationally offset from each other about the cutting member longitudinal axis (A) by 180° .

30. The cutting tool (20), according to claim 13, wherein: the two circumferential outer portions (48a, 48b) comprise: a rotationally preceding circumferential outer portion (48a) extending from the circumferential inner portion (50) in the rotationally preceding direction (D.sub.P); and a rotationally succeeding circumferential outer portion (48b) extending from the circumferential inner portion (50) in the rotationally succeeding direction (D.sub.S); wherein: the rotationally preceding circumferential outer portion (48a) and the rotationally succeeding circumferential outer portion (48b) intersect each other distal the circumferential inner portion (50).

31. The cutting tool (20), according to claim 13, wherein: the cutting member (22) is releasably clamped in the clamping recess (68) of the clamping sleeve adaptor (60) of the holding device (58); and the alignment indicator (76) is detachable from the clamping sleeve adaptor (60).
